

Birkhoff Stability Is Not Isometric Stability: Fixed-Vector Residual Transport for Hyper-Connections

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Abstract

Hyper-Connections expand residual connections into multiple residual streams, making residual transport itself a learnable geometric object. This creates a final-mile problem for residual architectures: unconstrained multi-stream transport is expressive but can destroy the identity mapping and amplify signals through depth, while Birkhoff-constrained transport, as used in manifold-constrained Hyper-Connections (mHC), restores the residual mean but can dissipate the mean-zero stream complement. We study this gap and propose IsoHC, a fixed-vector isometric residual transport constraint $H^\top H = I$, $H\mathbf{1} = \mathbf{1}$. IsoHC preserves the residual mean exactly while keeping every singular value on $\mathbf{1}^\perp$ equal to one. We show theoretically that mean preservation does not imply complement preservation: doubly stochastic transport can be perfectly stable in the residual mean direction while contracting $\mathbf{1}^\perp$ exponentially through depth. In residual-only detectors, unconstrained transport explodes, Birkhoff-like diffusion collapses the mean-zero energy, and IsoHC preserves gradients, energy, and stream diversity through 1024 layers. In a 48-layer FineWeb-Edu Transformer posthoc analysis, mHC’s composite mean-zero transport gain falls to 4.39×10^{-4} , whereas IsoHC remains near 0.996. At the current compute scale, we do not claim a decisive language-modeling perplexity improvement. Instead, we identify and repair a concrete residual-transport pathology: preserving the residual mean is not sufficient for preserving multi-stream residual information.

1 Introduction

Residual connections stabilize deep neural networks by preserving an identity path. A standard residual block can be abstracted as

$$x_{l+1} = x_l + F_l(x_l), \quad (1)$$

so that each layer writes an increment into a persistent residual state. Pre-normalized Transformers inherit this design: attention and MLP branches read from a shared residual stream and add updates back into it. This identity path is one of the main reasons very deep networks can be optimized.

Hyper-Connections (HC) generalize this paradigm by replacing the single residual stream with multiple residual streams and a learnable transport between them [3]. This turns the residual pathway from a scalar identity channel into a structured multi-stream transport system. However, it also makes the residual connection itself a geometric object. A generic learned transport matrix may amplify, collapse, or drift away from the residual identity semantics, and products of such matrices can become unstable through depth.

mHC repairs this instability by projecting residual mixing matrices onto the Birkhoff polytope [4]:

$$\mathcal{B}_n = \{H \in \mathbb{R}^{n \times n} : H\mathbf{1} = \mathbf{1}, \mathbf{1}^\top H = \mathbf{1}^\top, H \geq 0\}. \quad (2)$$

This is a natural stabilization: doubly stochastic transport preserves the uniform stream mean, restoring the identity-mapping property that residual connections are meant to provide. Recent work has further improved this line through alternative parameterizations and spectral constraints [5, 6].

Our observation is that mean preservation is not the same as residual-complement preservation. Let

$$P_\perp = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \quad (3)$$

be the projection onto the mean-zero stream subspace $\mathbf{1}^\perp$. In a multi-stream residual architecture, the stream mean carries the ordinary identity path, while $P_\perp X$ carries the differences between streams. If these differences are intended to store persistent residual information, then a residual transport geometry should not only preserve the mean; it should also avoid dissipating the complement.

Birkhoff geometry does not guarantee this. A doubly stochastic matrix is non-expansive, but it is generally diffusive. It can make all standard mean-direction stability diagnostics look healthy while exponentially contracting $\mathbf{1}^\perp$. We call this distinction:

Birkhoff stability is not isometric stability.

We propose IsoHC, a fixed-vector isometric constraint for residual transport:

$$\mathcal{M}_1 = \{Q \in O(n) : Q\mathbf{1} = \mathbf{1}\}. \quad (4)$$

IsoHC preserves the residual mean exactly and acts orthogonally on $\mathbf{1}^\perp$. It therefore avoids both unconstrained amplification and Birkhoff-style diffusion. The contribution of this note is not a claim of large-scale language-modeling superiority. Rather, it is a mechanism diagnosis and repair: mHC protects the residual mean, but it can dissipate the residual complement; fixed-vector isometry protects both.

Contributions. We make four contributions.

1. We formulate the residual-complement gap in multi-stream residual transport: preserving the stream mean does not preserve the full residual transport geometry.
2. We prove that Birkhoff constraints can hide depth-wise contraction on $\mathbf{1}^\perp$, even when mean-direction stability is exact.
3. We derive IsoHC as the fixed-vector orthogonal stabilizer that preserves both the residual mean and the complement norm.
4. We provide controlled evidence from residual-only detectors, graph oversmoothing experiments, and 48-layer Transformer posthoc analysis.

2 Multi-Stream Residual Transport

Let $X_l \in \mathbb{R}^{n \times d}$ denote n residual streams at layer l . A generic multi-stream residual update can be written as

$$X_{l+1} = H_l X_l + b_l \otimes y_l, \quad (5)$$

where $H_l \in \mathbb{R}^{n \times n}$ transports the streams and $y_l \in \mathbb{R}^d$ is the branch output. The stream mean is

$$\mu(X) = \frac{1}{n} \mathbf{1}^\top X. \quad (6)$$

To embed the standard residual path into the mean stream, we want

$$\mu(X_{l+1}) = \mu(X_l) + y_l. \quad (7)$$

This is satisfied if

$$\mathbf{1}^\top H_l = \mathbf{1}^\top, \quad \frac{1}{n} \mathbf{1}^\top b_l = 1. \quad (8)$$

Every stream state decomposes as

$$X = \mathbf{1} \otimes \mu(X) + P_\perp X. \quad (9)$$

The first term is the residual mean. The second term is the residual complement, where the additional degrees of freedom of a multi-stream residual architecture live. This gives two distinct requirements:

$$\text{mean direction:} \quad \text{preserve the residual identity path,} \quad (10)$$

$$\text{mean-zero complement:} \quad \text{transport stream differences without expansion or dissipation.} \quad (11)$$

mHC solves the first requirement by using Birkhoff constraints. IsoHC targets both.

3 Birkhoff Stability Versus Complement Preservation

Let $U \in \mathbb{R}^{n \times (n-1)}$ be an orthonormal basis of $\mathbf{1}^\perp$, so that $U^\top U = I$ and $U^\top \mathbf{1} = 0$. The complement action of a mean-preserving matrix H is

$$H_\perp = U^\top H U. \quad (12)$$

Theorem 1 (Birkhoff transport is non-expansive, not necessarily isometric). *If $H \in \mathcal{B}_n$, then H preserves the stream mean and*

$$\|U^\top H U\|_2 \leq 1. \quad (13)$$

However, Birkhoff constraints do not imply that $U^\top H U$ has singular values equal to one.

Proof. By the Birkhoff-von Neumann theorem, H is a convex combination of permutation matrices:

$$H = \sum_k \alpha_k P_k, \quad \alpha_k \geq 0, \quad \sum_k \alpha_k = 1. \quad (14)$$

Each P_k is orthogonal, hence $\|P_k\|_2 = 1$. Therefore

$$\|H\|_2 \leq \sum_k \alpha_k \|P_k\|_2 = 1. \quad (15)$$

Since $H\mathbf{1} = \mathbf{1}$ and $\mathbf{1}^\top H = \mathbf{1}^\top$, the subspace $\mathbf{1}^\perp$ is invariant under H , and the restriction $U^\top H U$ is also non-expansive. Non-expansiveness does not force all singular values to equal one. $\square$

Theorem 2 (Mean stability can hide exponential complement contraction). *For $0 \leq \alpha < 1$, define*

$$H_\alpha = \alpha I + (1 - \alpha) \frac{\mathbf{1}\mathbf{1}^\top}{n}. \quad (16)$$

Then $H_\alpha \in \mathcal{B}_n$, $H_\alpha \mathbf{1} = \mathbf{1}$, and for any $x \in \mathbf{1}^\perp$,

$$H_\alpha x = \alpha x. \quad (17)$$

Consequently,

$$\|P_\perp H_\alpha^L P_\perp x\|_2 = \alpha^L \|P_\perp x\|_2. \quad (18)$$

Proof. H_α is nonnegative for $0 \leq \alpha < 1$, and its rows and columns sum to one, so $H_\alpha \in \mathcal{B}_n$. If $x \in \mathbf{1}^\perp$, then $\mathbf{1}^\top x = 0$ and $\mathbf{1}\mathbf{1}^\top x/n = 0$. Thus $H_\alpha x = \alpha x$, and the depth- L statement follows by repeated application. $\square$

Theorem 2 is the core counterexample: row/column stochasticity can be perfect while complement information decays exponentially. This is not a bug in mHC. mHC is designed to restore identity stability. The point is that identity stability alone does not certify multi-stream complement preservation.

4 IsoHC: Fixed-Vector Isometric Transport

IsoHC replaces Birkhoff residual transport with the fixed-vector orthogonal stabilizer

$$\mathcal{M}_1 = \{Q \in O(n) : Q\mathbf{1} = \mathbf{1}\}. \quad (19)$$

Theorem 3 (Fixed-vector isometry preserves mean and complement norm). *Let $Q \in \mathcal{M}_1$. Then for any $X \in \mathbb{R}^{n \times d}$,*

$$\mu(QX) = \mu(X), \quad (20)$$

and

$$\|P_\perp QX\|_F = \|P_\perp X\|_F. \quad (21)$$

Equivalently, all singular values of $U^\top Q U$ are one.

Proof. Since $Q^\top Q = I$ and $Q\mathbf{1} = \mathbf{1}$, we also have $Q^\top \mathbf{1} = \mathbf{1}$. Hence

$$\mu(QX) = \frac{1}{n} \mathbf{1}^\top QX = \frac{1}{n} \mathbf{1}^\top X = \mu(X). \quad (22)$$

The subspace $\mathbf{1}^\perp$ is invariant under Q , and the restriction $U^\top QU$ is orthogonal. Therefore it preserves the Frobenius norm of the complement. $\square$

Proposition 1 (Nonnegative orthogonal stochastic matrices are discrete). *If $Q \geq 0$, $Q\mathbf{1} = \mathbf{1}$, and $Q^\top Q = I$, then Q is a permutation matrix.*

Proof. The rows of Q are nonnegative unit vectors with pairwise inner product zero. Thus distinct rows have disjoint support. Since each row sums to one and has unit Euclidean norm, each row has exactly one nonzero entry, equal to one. Column orthogonality then implies that each column has exactly one such entry. $\square$

This proposition explains why continuous non-diffusive residual transport must leave the nonnegative Birkhoff geometry. If we require nonnegativity, stochasticity, and orthogonality simultaneously, the feasible set degenerates to permutations.

4.1 Projection and finite-step retraction

Let $v = \mathbf{1}/\sqrt{n}$ and let U span $v^\perp$. Every element of $\mathcal{M}_1$ has the form

$$Q = vv^\top + URU^\top, \quad R \in O(n-1). \quad (23)$$

Given an unconstrained matrix A , the Frobenius-nearest exact projection, away from polar degeneracies, is

$$\Pi_v(A) = vv^\top + U \text{polar}(U^\top AU) U^\top. \quad (24)$$

In implementation, we use a finite-step Newton-Schulz polar iteration as an efficient fixed-vector retraction:

$$B = U^\top AU, \quad R_K \approx \text{polar}(B), \quad Q_K = vv^\top + UR_K U^\top. \quad (25)$$

The finite-step retraction preserves the fixed vector by construction, but orthogonality is approximate and must be monitored. In our 48-layer run, IsoHC’s fixed-vector error is 2.79×10^{-7} and its orthogonality error is 5.57×10^{-4} .

5 Experimental Evidence

The experiments are designed to test residual transport geometry, not to claim language-modeling state of the art. The central question is:

Does the transport preserve both the residual mean and the residual complement through depth?

5.1 Residual-only depth detector

We first remove attention and MLP branches and study pure transport through up to 1024 layers. Table 1 shows the $n = 8$, $L = 1024$ detector.

The detector isolates the geometry. Unconstrained HC can amplify through depth; Birkhoff-like diffusion prevents explosion but collapses the complement; IsoHC preserves the complement isometrically.

5.2 48-layer FineWeb-Edu mechanism run

We trained a 48-layer GPT-style Transformer on a FineWeb-Edu token cache:

$$L = 48, \quad d_{\text{model}} = 512, \quad h = 8, \quad n = 4, \quad T = 512, \quad (26)$$

for 20M tokens per method. The model has roughly 177M parameters. A fair batch probe selected a common batch size across methods.

Table 1: Residual-only transport detector at $n = 8$, $L = 1024$. IsoHC preserves gradient scale, mean-zero energy, and stream diversity. Diffusive transport collapses the complement, while unconstrained transport explodes.

method	grad mean	mean-zero energy	stream cosine	$\sigma_{\min}$	$\sigma_{\max}$
IsoHC	1.0000	1.0000	0.0250	1.0000	1.0000
mHC-lite	0.3536	0.0000	1.0000	0.5724	1.0000
GCN diffusion	0.3533	0.0000	1.0000	0.8000	1.0000
unconstrained	5.58×10^{13}	5.61×10^{13}	0.9877	0.4270	1.6155

Table 2: 48-layer FineWeb-Edu short-budget run. Perplexity differences are not the main claim. The key observation is that mHC reduces $E_{\perp}$, while IsoHC preserves much larger complement energy.

method	val loss	val PPL	$E_{\perp}$ init $\rightarrow$ final	stream cosine init $\rightarrow$ final
baseline	5.7366	310.01	–	–
identity-HC	5.7157	303.60	0.0222 $\rightarrow$ 0.0503	0.9993 $\rightarrow$ 0.9971
unconstrained HC	5.6760	291.78	0.0222 $\rightarrow$ 0.0547	0.9993 $\rightarrow$ 1.0000
mHC	5.7249	306.39	0.0224 $\rightarrow$ 0.0136	0.9993 $\rightarrow$ 0.9998
IsoHC	5.7206	305.10	0.0224 $\rightarrow$ 0.0546	0.9993 $\rightarrow$ 0.9967

5.3 Checkpoint posthoc intervention

We further analyzed checkpointed mHC and IsoHC runs using 16 validation batches with batch size 1 for intervention and gradient probes.

The posthoc result is the cleanest Transformer mechanism evidence: mHC’s composite complement transport nearly vanishes, while IsoHC remains isometric. However, complement-removal and IsoHC-to-identity replacement change average validation loss only at the 10^{-4} to 10^{-3} scale. This means the current run establishes the transport pathology, but does not yet show that the preserved complement is strongly used by the language model.

5.4 Graph oversmoothing side evidence

Graph neural networks provide an additional stress test for diffusion. In a synthetic oversmoothing experiment at depth 128, ordinary diffusion collapses node variation, while an isometric transport preserves it:

On Cora node classification, a corrected stream architecture recovers deep performance where ordinary GCNs collapse:

6 Discussion

What IsoHC solves. IsoHC should not be read as a small-model perplexity trick. It targets residual transport geometry. Ordinary residual connections provide a stable identity path but only one residual transport channel. HC adds multiple streams, but unconstrained transport can destabilize the identity path. mHC restores the mean identity path with Birkhoff geometry. IsoHC completes this geometry by also preserving the residual complement.

Why not just identity-HC? The identity matrix is a member of $\mathcal{M}_1$ and preserves both mean and complement. It is therefore an essential control. IsoHC’s additional value is learnable non-destructive stream communication: $U^{\top} H_l U$ can be a nontrivial orthogonal transform rather than the identity. Our current LM replacement probe shows only a small IsoHC-to-identity loss delta, so the present evidence verifies the geometry but does not yet demonstrate strong functional value of learned rotations over identity-HC.

Table 3: Single-layer transport diagnostics in the 48-layer run. mHC satisfies stochastic constraints but has average complement singular value near 0.922. IsoHC keeps the fixed vector and complement singular values near one.

method	invariant error	orthogonality error	$\mathbf{1}^\perp$ singular values
identity-HC	0.0	0.0	exact identity
unconstrained HC	0.1050	0.1694	unconstrained
mHC	row 0.0046 / col 2.30×10^{-7}	—	mean 0.9222, min 0.9208, max 0.9237
IsoHC	2.79×10^{-7}	5.57×10^{-4}	mean 0.99996, min 0.99982, max 1.00012

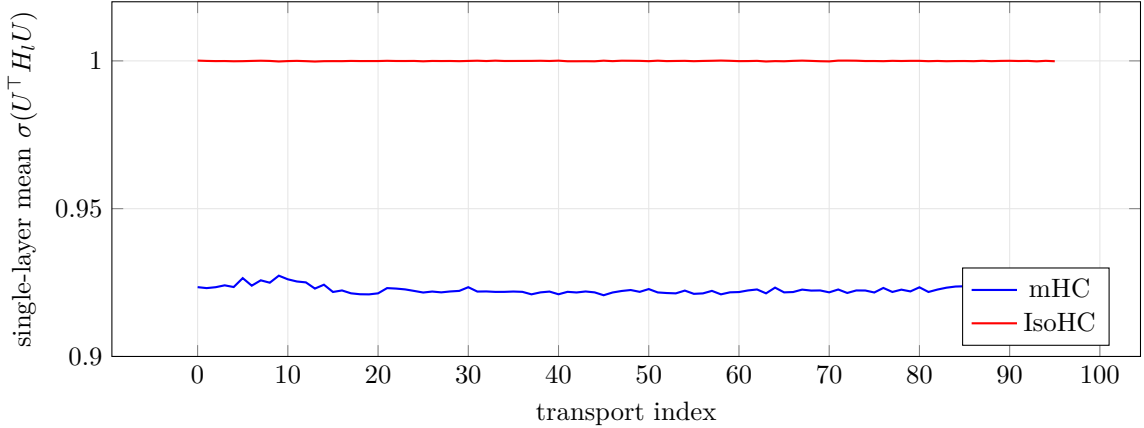


Figure 1: Single-layer complement singular values in the 48-layer Transformer. mHC is consistently contractive on $\mathbf{1}^\perp$, while IsoHC remains near one.

Relation to spectral HC. Spectral constraints such as $\|H\|_2 \leq 1$ prevent expansion but do not by themselves guarantee residual mean preservation or complement isometry. IsoHC imposes the stronger geometry

$$H\mathbf{1} = \mathbf{1}, \quad H^\top H = I, \quad (27)$$

which simultaneously preserves the residual mean and all complement singular values. Thus the distinction is not simply signed versus nonnegative mixing, but non-expansive versus isometric transport on the invariant complement.

7 Limitations

This note intentionally makes a limited claim.

1. We do not show decisive large-scale LM perplexity improvements over mHC or spectral HC variants.
2. The current LM complement-intervention effects are small, so functional use of the preserved complement remains weak at this scale.
3. Identity-HC remains a strong control. Future work must show when learned isometric stream mixing is more useful than no mixing.
4. IsoHC gives fixed-vector isometry for the residual transport component, not full dynamical isometry of an entire Transformer block.
5. Finite-step Newton-Schulz retractions are approximate; fixed-vector error and orthogonality error must be monitored.

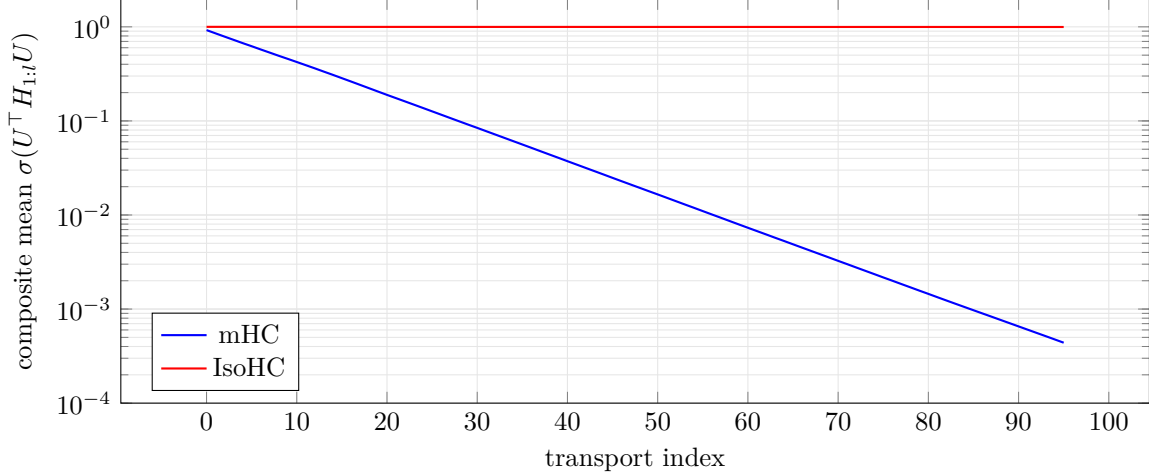


Figure 2: Composite complement transport gain. At the final transport, mHC falls to 4.39×10^{-4} , whereas IsoHC remains near 0.996. This is the central mechanism result: mean-preserving stability does not certify complement-preserving transport.

Table 4: 48-layer checkpoint posthoc analysis. The transport pathology is strong, but functional loss deltas are small at this budget.

method	val loss	PPL	final composite $\mathbf{1}^\perp$ gain	best removal ΔL	replace by identity ΔL
mHC	5.900727	365.30	0.000439	0.000289	-0.000524
IsoHC	5.901274	365.50	0.996349	0.000516	0.000713

8 Future Work

The next experiments should test whether the preserved complement is functionally used, rather than merely energetically preserved.

Token-level intervention. For each token, compare full logits p_t with intervention logits p'_t after complement removal or complement clamping:

$$D_{\text{KL}}(p_t \| p'_t), \quad \Delta \ell_t = -\log p'_t(y_t) + \log p_t(y_t). \quad (28)$$

The average loss delta may be small even if high-entropy, rare, late-position, or copy-sensitive tokens depend on the complement.

Complement clamp and gate sweeps. Instead of removing the complement at one layer, clamp it from layer k onward:

$$X_l \leftarrow \mathbf{1}\mu(X_l), \quad l \geq k, \quad (29)$$

or sweep

$$X_l^{(\alpha)} = \mathbf{1}\mu(X_l) + \alpha P_\perp X_l, \quad \alpha \in \{0, 0.25, 0.5, 0.75, 1\}. \quad (30)$$

Monotonic sensitivity to α would provide a stronger causal bridge.

Deep-thin stress and routing tasks. Depth is the natural axis for a contraction claim. Future work should run 48/72/96/128-layer deep-thin models and synthetic routing or associative recall tasks that require cross-stream communication. These settings can test whether learned isometric mixing is functionally necessary and whether mHC’s complement contraction limits behavior.

Table 5: Synthetic graph oversmoothing stress test at depth 128.

method	energy	centered variance	cosine	Dirichlet energy
GCN	0.2003	0.0012	1.0000	0.0000
Residual GCN	0.2003	0.0012	1.0000	0.0000
IsoNode	1.0000	1.0000	0.0325	7940.2

Table 6: Cora deep node classification. This is side evidence for the anti-collapse role of isometric stream transport, not the main LM claim.

model	L2	L16	L32
GCN	77.6 ± 1.9	31.1 ± 0.2	31.2 ± 0.3
ResGCN	75.8 ± 1.2	28.8 ± 4.0	28.6 ± 2.8
IsoStream v2	–	73.1 ± 1.9	66.6 ± 3.4

9 Conclusion

Hyper-Connections make residual transport multi-stream. mHC restores the residual mean by imposing Birkhoff constraints, but preserving the mean is not the same as preserving the full residual transport geometry. Birkhoff transport is non-expansive and can be strongly contractive on the mean-zero complement. IsoHC imposes fixed-vector isometry, preserving both the residual mean and the complement norm. The current evidence supports IsoHC as a geometric mechanism: mHC’s composite complement transport collapses in depth, while IsoHC remains approximately isometric. The remaining open question is functional relevance: when, and at what scale, does the preserved residual complement become necessary for model behavior?

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